HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered structural calculations and design documentation for temporary works solutions, including propping and shoring systems, for a regional infrastructure delivery programme.
- Prepared detailed technical reports and drawings for complex temporary works designs, ensuring compliance with project-specific requirements and construction methodologies.
- Collaborated with senior engineers and site teams to develop and refine practical, buildable temporary works designs, responding to technical queries during the construction phase.
- Reviewed permanent works designs and site information to inform the development of appropriate temporary works solutions, focusing on excavation and working platform stability.
- Developed Python scripts to automate the generation of engineering calculation summaries, reducing manual reporting time by approximately 30% for a team of 8 engineers.
- Contributed to the design checks and technical reviews of temporary works proposals, ensuring adherence to relevant industry standards and health and safety regulations.
- Containerised a Python-based structural analysis tool using Docker for consistent deployment across development and testing environments.

SKILLS

Engineering Design – Temporary Works Design, Structural Engineering Principles, Engineering Calculations, Technical Reports, Drawings, Propping, Shoring, Formwork, Falsework, Working Platforms, Temporary Stability, Excavations

Software & Tools – AutoCAD, Structural Design Software, Python, Docker, Git, Jira, Confluence

Project Management – Design Documentation, Technical Documentation, Project Records, Site Inspections, Technical Queries, Health and Safety Requirements

Domain Knowledge – Construction Projects, Civil Engineering, Structural Engineering, Multidisciplinary Project Team

Core Skills – Problem-Solving, Communication Skills, Collaboration

PROJECTS

Temporary Works Design for Bridge Abutment Stabilization

- Developed detailed design calculations and drawings for temporary propping and shoring systems to support bridge abutments during construction.
- Created comprehensive technical documentation outlining the methodology and safety protocols for the temporary works installation, reviewed by project stakeholders.
- Utilized structural analysis software to validate the load-bearing capacity and stability of the designed temporary works under various site conditions.

Automated Structural Load Calculation Script

- Engineered a Python script to automate the calculation of complex structural loads for temporary works scenarios, processing input parameters from standardized templates.
- Integrated the script with basic error-handling and output formatting, reducing the potential for manual input errors by over 40% in a pilot test.
- Documented the script's usage and functionality, enabling other engineers on the team to leverage it for routine calculations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)